

The Pleistocene-Holocene Transition in Tagua-Tagua, central Chile

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Abstract

Paleohydrological reconstructions from the Chilean Altiplano document abrupt moisture fluctuations during the last millennia. Although the end of the mid Holocene aridity and the onset of more humid conditions between 6–4 ka has been identified in numerous regional marine and terrestrial sites, the timing of late Holocene dry and humid phases shows large regional variability. Laguna Miscanti and Laguna Miñiques (23° 43'S – 67° 46'W, 4140 m asl) are topographically closed, but connected by surface outflow from Miscanti. Sedimentological and geochemical indicators from two new cores show large facies changes, i.e. higher carbonate and evaporite deposition during more arid periods and increased organic productivity (both algal and macrophyte) during more humid phases. As in most Andean lakes located in volcanic settings, large 14C reservoir effects occur complicating 14C dating, so the age models include 210Pb and U/Th dating. In spite of dating uncertainties, both lakes show similar patterns during the last millennium. A humid phase in Laguna Miscanti prior to ca 1200 CE is coherent with rodent middens and geomorphological features indicative of a major pluvial/recharge event at lower altitudes (Atacama Desert) during the Medieval Climate Anomaly (ca 800 - 1300 CE). The LIA (1300 – 1850 CE) is characterized by several arid/humid cycles and the last century by a productivity increase. The hydrological changes observed during the last millennium illustrate the complex dynamics of recent climate evolution over the high altitude Andean plateau. Discrepancies between the northern and southern Altiplano records and with intermediate latitudes (Central Chile) records may reflect contrasting responses to external forcing (Westerlies versus South American Monsoon dynamics) along different climatic zones.

Key words: keywords

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16 1. Introduction

17 2. Study site and regional setting

18 3. Materials and methods

19 In November 2019, we recovered a sedimentary sequence of 15 cm length,
20 10 cm width and 280 cm depth of the profile exposed on the western wall of a
21 1x2 m excavation in the ALTT (Tagua-Tagua 3, Figure 1). Sediment samples
22 (5 to 10 g of material) were obtained at 1-cm intervals. Sixty-six samples
23 were sent to the laboratories of Pyrenees Ecology Institute (IPE-CSIC) and
24 the Andalusian Institute Earth Sciences Institute (IACT-CSIC) in Spain for
25 geochemical, pollen and isotopic analysis. Another set of samples were sent to
26 the Archaeobotany and Palaeoecology Laboratory of the University of Exeter
27 for the processing and extraction of diatom and phytolith. Sedimentary facies
28 were defined and characterized based on macroscopic description (color, bedding
29 features, grain size, lithology and sedimentary textures) and a machine learning
30 workflow using the scikit-learn library (cite) in Python. Sedimentary facies
31 were defined using a dimensionality reduction (Exploratory Factor Analysis) and
32 unsupervised machine learning (K-Means algorithm cluster) with three different
33 dataset (elemental geochemistry, grain size and diatoms)(Hall., 2016).

34 Samples were taken every 2 cm for analysed of total carbon (TC), total
35 inorganic carbon (TIC), total organic carbon ($TOC = T - TIC$), total nitrogen
36 (TN) and total sulfur (TS) in a LECO SCN furnace (IPE-CSIC). The grain-
37 size frequency distributions (between 0.02 μm and 2000 μm) were measured in
38 triplicate using a Malvern Mastersizer 2000 APA2000 grain-size analyzer with
39 Hydro 2000G AWA2000. To determine the grain-size, the samples were pre-
40 treated with 10% H_2O_2 and 10% HCl to remove organic matter and carbonates,
41 respectively. Robust End-member mixing analysis (EMMA) was performed
42 using the generic deterministic EMMA protocol with the R package EMMAgeo
43 v0.9.8. (Dietze et al 2012; Dietze and Dietze 2019) to infer the sediment sources,
44 transport and depositional processes. Stable carbon and oxygen isotope analyses
45 on calcite were carried out every 2 cm by mass spectrometry in bulk sediment
46 samples following standard procedures and using an IRMS Finnigan MAT 251.
47 The isotopic values are reported in the conventional delta notation relative to
48 the PDB standard.

49 Eleven pollen samples were processed following Faegri and Iversen (1989)
50 methods to extract pollen grains. Lycopodium tablets were added for calculating
51 pollen concentration (grains/ cm^3 ; Stockmarr(1971)) and accumulation rates
52 (grains/ cm^2/yr). Pollen grains were mounted on glass slides and identified and
53 quantified under an Axiostar Carl Zeiss microscope (400x and 1000x) using
54 published pollen atlas Heusser (1971) and the reference pollen collection of the
55 Paleocology Laboratory at IPE-CSIC. Pollen counts include 300 terrestrial
56 pollen grains excluding paludal taxa. Phytoliths were obtained from twenty
57 four samples of approximately 2 gr of sediment. We followed the standard
58 procedures for phytoliths extraction (Piperno, 1988, 1994, 2006) using 36%
59 HCl and 70% HNO_3 to remove carbonates and organic matter respectively.

60 Phytoliths were separated by heavy liquid flotation using ZnBr₂ prepared to a
61 density of 2.3 g/cm³ (Dickau et al., 2013). Finally, phytoliths were counted until
62 obtaining a total of 200 per sample and they were classified according to the
63 International Code for Phytolith Nomenclature (Madella et al. 2005). Diatoms
64 were extracted simultaneously with phytoliths. Digested aliquotes were mounted
65 onto permanent slides with Naphrax. Diatom valves were visualized under a
66 XXXXX microscope, with an oil immersion objective of 1000x. A minimum of
67 400 valves were counted for each level, with the exception of the base levels X,
68 Y, Z, where valves were scarce. Diatoms were identified to species level with
69 the LM images (1000x) taken using a digital SLR camera (Canon EOS Rebel)
70 attached to microscope AND SEM???. The following literature was used for
71 taxonomy and autoecology of the encountered valves.

72 The TT3 chronology was reconstructed with thirteen AMS-radiocarbon
73 dates on charcoal and bulk sediment determined at DirectAMS laboratories,
74 USA. Results are presented in units of percent modern carbon (pMC) and the
75 uncalibrated radiocarbon age before present (relative to 1950 CE). Radiocarbon
76 ages were calibrated using the Southern Hemisphere calibration curve (SHCal20,
77 Hogg et al 2020). We used the package rbacon v2.2 (Blaauw and Christen, 2011)
78 for age–depth modeling and to establish the deposition rates along the sequence.
79 A change point analysis was carried out to identify significant breaks in the time
80 series (Bai and Perron, 2003; Killick and Eckley, 2014). Stratigraphic diagrams
81 were analyzed and plotted using ‘tidypaleo’ v0.1.1, ‘ggplot’ v3.3.5 and ‘rioja’
82 v0.9-26 packets in R 4.1.0 software.

4. Results

4.1. Sedimentary facie, geochemical and end-member mixing analysis

4.2. Phytoliths and Pollen analysis

4.3. Diatom analysis

4.4. Radiocarbon chronology

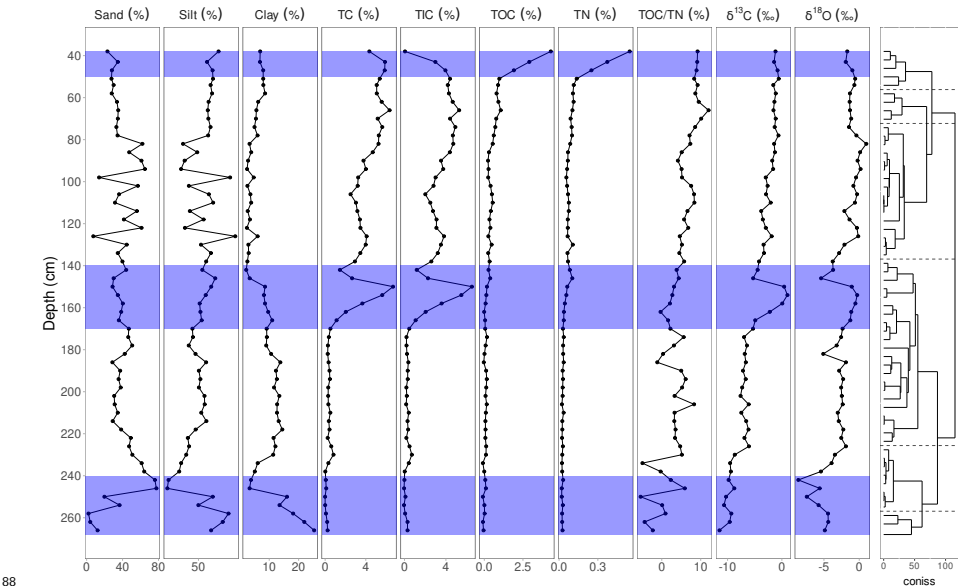


Table 1: Radiocarbon dating of TT3 sequence. The table shows the ages in terms of years 'before present' (BP), percent modern carbon (pMC) and their uncertainties (1σ). The median age calibration was conducted with Oxcal 4.4 and SHCal20 calibration curve (Hogg et al., 2020; Bronk Ramsey 2009a)

Lab code	Depth	^{14}C a BP	1σ	pMC	1σ	Median a cal BP	1σ	Material
D-AMS 039542	39	5345	28	51.41	0.18	6086	69	Charcoal
D-AMS 039543	44	5806	33	48.54	0.20	6571	56	Charcoal
D-AMS 039544	67	7259	32	40.51	0.16	8021	62	Charcoal
D-AMS 039545	106	8657	45	34.04	0.19	9591	57	Charcoal
D-AMS 039546	150	9736	35	29.76	0.13	11134	111	Bulk sediment
D-AMS 039547	161	10211	63	28.05	0.22	11817	144	Bulk sediment
D-AMS 039548	162	10936	41	25.63	0.13	12813	46	Charcoal
D-AMS 039549	162	10747	49	26.24	0.16	12709	38	Bulk sediment
D-AMS 039550	168	10590	39	26.76	0.13	12549	64	Charcoal
D-AMS 039551	199	11117	51	25.06	0.16	13015	61	Bulk sediment
D-AMS 039552	223	9977	45	28.88	0.16	11377	122	Bulk sediment
D-AMS 039553	254	17353	84	11.53	0.12	20890	124	Bulk sediment
D-AMS 039554	266	16558	69	12.73	0.11	19959	130	Bulk sediment

NULL

⁹⁰ **5. Discussions**

⁹¹ **References**

Table 2: Sedimentological, compositional characteristics and depositional environment of Tagua-Tagua lacustrine facies.

Banded diatomaceous silts (TOC <2%; BioSi 18-19 %)			
Facies	Sedimentological properties	Composition and geochemistry	Depositional environment
S1	Brown, fine sand (25 %) to fine silts (40 %) with interbedded gravel clasts and modern edaphic features	High TOC (3.1 %) and TN (0.5 %) and low TIC (0.01 %) and TS (0.05 %). High $\delta^{13}C_{CO_2}$ and $\delta^{18}O_{CO_2}$ values (-21 ‰ and -11 ‰, respectively)	Modern soil
D2	Dark grey to brown, diatom-aceous, medium to coarse silt in cm-thick layers with gradational top and bottom boundaries.	Lower TOC (1.6 %), TIC (< 0.1), TS (0.22 %) and BioSi (18.2 %) than facies D1. High silicate-related elements (K, Ti, Rb, Sr and Zr); low Br/Ti, Ca/Ti, S/Ti and Fe/Mn.	Distal, profundal environments with higher clastic - volcanic input
Banded, diatomaceous, organic-rich silts (TOC >3%; BioSi > 20%)			
Facies	Sedimentological properties	Composition and geochemistry	Depositional environment
D3	Banded, dm-thick layers of dark brown diatomaceous, organic rich medium to coarse silts with variable macrophyte content.	High TOC (3.8 %) and BioSi (21.7 %), moderate TS (0.6 %) and low TIC (<0.1 %). Relatively high Si, Fe and Br and low Ti, Rb, Sr, Zr, Ti, Br/Ti, Ca/Ti, Fe/Mn.	Littoral to relatively deep environments dominated by organic deposition with low clastic input. Sequences (D3/D5,D4), (D5/D3/D4) and alternation D3/D4 represent transition from littoral, macrophyte-dominated (D3 and D5) to deeper environments (D4).
D4	Massive to banded, < dm-thick layers of lighter brown diatom and organic rich fine silt.	High TOC (3.2%) and BioSi (21.4 %), moderate TS (0.7 %) and low TIC (< 0.1 %). Relatively high Si, Fe and Br and low Ti, Rb, Sr, Zr.	
D5	Massive to banded, mm- to cm - thick layers of brown, macrophyte-rich coarse silt.	TS (1.4 %) and presence of calcite (TIC = 0.1%). Higher TOC/TN. Moderate but fluctuating silicate-related elements (K, Ti, Rb, Sr and Zr), high Br/Ti, Ca/Ti, and S/Ti. High Mn values, and low Fe/Mn ratios.	
Laminated diatomaceous organic-rich silts(TOC 2-3.5%; BioSi ca. 20 %)			
Facies	Sedimentological properties	Composition and geochemistry	Depositional environment
D6	Laminated, dark brown, organic-rich silt composed of three main laminae type: a) greenish grey, diatomaceous medium to fine silt; b) dark brown, organic-rich fine silt with diffuse upper and lower boundaries and c) brown, macrophyte-rich, coarse silt, with sharp, erosional lower boundaries. Facies D6b contains more frequent endogenic calcite crystals.	High TOC (3.1%), BioSi (19.7%) and TS (1.4%, peaks up to 6% in D6c) and common presence of calcite (TIC >= 0.1). Moderate but fluctuating silicate-related elements (K, Ti, Rb, Sr and Zr); high Br/Ti, Ca/Ti, and S/Ti. High Mn values and low Fe/Mn ratios.	Distal, frequently oxic environments dominated by organic deposition and higher clastic input. Some periods with endogenic calcite formation